# Lecture 10: LTI Resistive Circuits and Equivalent One-Ports

## EE201 – Circuit Theory I

M. Mert Ankaralı

Department of Electrical and Electronics Engineering
Middle East Technical University

current voltage a part of a circuit (box $N$)

# Review: equations from the graph alone

| | **incidence / mesh** | **fundamental cutset / loop** |
|---|---|---|
| KCL | $A\mathbf{i} = \mathbf{0}$ | $Q\mathbf{i} = \mathbf{0}$ |
| KVL | $M\mathbf{v} = \mathbf{0}$ | $B\mathbf{v} = \mathbf{0}$ |
| $n-1$ voltages | node voltages $\mathbf{e}$ | tree voltages $\mathbf{v}_T$ |
| $b-n+1$ currents | mesh currents $\mathbf{i}_M$ | co-tree currents $\mathbf{i}_C$ |
| works for | $M$: planar, nonseparable | any connected graph |

KCL and KVL depend only on the graph: they hold for any elements.

A circuit has $2b$ unknowns: $b$ more equations come from the elements.

# Why this lecture?

- **So far:** KCL and KVL, written for any elements.

## Why this lecture?

- **So far:** KCL and KVL, written for any elements.
- **Now:** LTI resistive circuits, where every equation is linear and algebraic.

## Why this lecture?

- **So far:** KCL and KVL, written for any elements.
- **Now:** LTI resistive circuits, where every equation is linear and algebraic.
- **Two ways to solve:** transform the circuit into a simpler one, or write its equations as it is.

## Why this lecture?

- **So far:** KCL and KVL, written for any elements.
- **Now:** LTI resistive circuits, where every equation is linear and algebraic.
- **Two ways to solve:** transform the circuit into a simpler one, or write its equations as it is.
- **This lecture — transform:** replace a part by a source, or by a simpler part with the same terminal behaviour
  - substitution theorem, equivalent one-ports
  - series and parallel connections, source transformation

# Why this lecture?

- **So far:** KCL and KVL, written for any elements.
- **Now:** LTI resistive circuits, where every equation is linear and algebraic.
- **Two ways to solve:** transform the circuit into a simpler one, or write its equations as it is.
- **This lecture — transform:** replace a part by a source, or by a simpler part with the same terminal behaviour
  - substitution theorem, equivalent one-ports
  - series and parallel connections, source transformation
- **Later — the circuit as it is:** node voltage and mesh current analysis

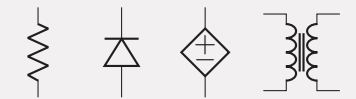
# Roadmap

1. **LTI resistive circuits** — what the words mean, and what we gain
2. **A first example** — simplify a circuit step by step
3. **Substitution theorem** — replace a part of a circuit by a source
4. **One-ports and equivalence** — the same $i$–$v$ characteristic at the port
5. **Series and parallel** — combine resistors and sources
6. **One source and one resistor** — exchange a voltage source and a current source

# Resistive and dynamic elements

## Resistive element: no memory

Its law relates $v$ and $i$ at the same instant, algebraically.

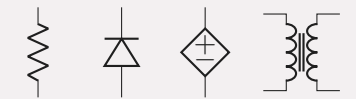



later: the ideal op-amp

# Resistive and dynamic elements

## Resistive element: no memory

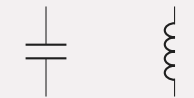
Its law relates $v$ and $i$ at the same instant, algebraically.




later: the ideal op-amp

## Dynamic element: memory

Its law has a derivative.

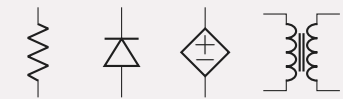



also coupled inductors

# Resistive and dynamic elements

## Resistive element: no memory

Its law relates $v$ and $i$ at the same instant, algebraically.




later: the ideal op-amp

## Dynamic element: memory

Its law has a derivative.



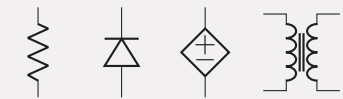

also coupled inductors

"Resistive" means memoryless, not only "a resistor".

# Resistive and dynamic elements

## Resistive element: no memory

Its law relates $v$ and $i$ at the same instant, algebraically.




later: the ideal op-amp

## Dynamic element: memory

Its law has a derivative.



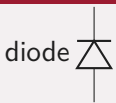

also coupled inductors

"Resistive" means memoryless, not only "a resistor".

An element is linear if its law is linear in $v$, $i$; time-invariant if its law does not change with time.

# Three questions about a circuit

## Linear?

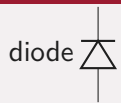



Nonlinear if it contains at least one nonlinear element, e.g. a diode.

Otherwise **linear**.

# Three questions about a circuit

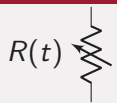
## Linear?




Nonlinear if it contains at least one nonlinear element, e.g. a diode.

Otherwise **linear**.

## Time-invariant?

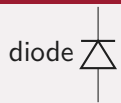



Time-varying if it contains at least one time-varying element, e.g. $R(t)$.

Otherwise **time-invariant**.

# Three questions about a circuit

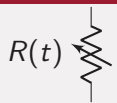
## Linear?




Nonlinear if it contains at least one nonlinear element, e.g. a diode.

Otherwise **linear**.

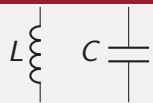
## Time-invariant?




Time-varying if it contains at least one time-varying element, e.g. $R(t)$.

Otherwise **time-invariant**.

## Resistive?

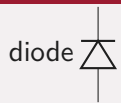



Dynamic if it contains at least one dynamic element: an $L$ or a $C$.

Otherwise **resistive**.

# Three questions about a circuit

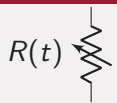
## Linear?




Nonlinear if it contains at least one nonlinear element, e.g. a diode.

Otherwise **linear**.

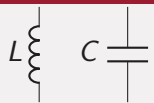
## Time-invariant?




Time-varying if it contains at least one time-varying element, e.g. $R(t)$.

Otherwise **time-invariant**.

## Resistive?



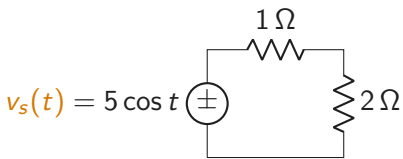

Dynamic if it contains at least one dynamic element: an $L$ or a $C$.

Otherwise **resistive**.

**LTI resistive circuit = linear + time-invariant + resistive**:

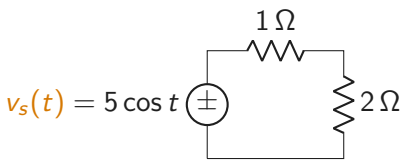
apart from the independent sources, every element is linear, time-invariant and resistive.

# Independent sources are inputs




The source varies in time. Is the circuit time-varying?

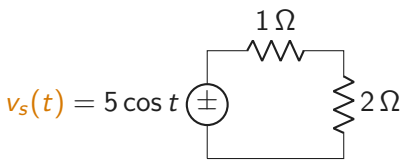
# Independent sources are inputs




The source varies in time. Is the circuit time-varying?

No. An independent source is an external input: a given signal of time.

# Independent sources are inputs




The source varies in time. Is the circuit time-varying?

No. An independent source is an external input: a given signal of time.

The three questions are about the circuit that the inputs drive, not about the inputs. This is an LTI resistive circuit.

## What varies in time?

In every circuit $v(t)$ and $i(t)$ are functions of time. The question is whether the element law changes with time.

**Time-invariant resistor**

$$v(t) = R\, i(t)$$

the same rule at every instant

## What varies in time?

In every circuit $v(t)$ and $i(t)$ are functions of time. The question is whether the element law changes with time.

**Time-invariant resistor**

$$v(t) = R\, i(t)$$

the same rule at every instant

**Time-varying resistor**

$$v(t) = R(t)\, i(t)$$

the rule itself changes with $t$

## What varies in time?

In every circuit $v(t)$ and $i(t)$ are functions of time. The question is whether the element law changes with time.

**Time-invariant resistor**

$$v(t) = R\, i(t)$$

the same rule at every instant

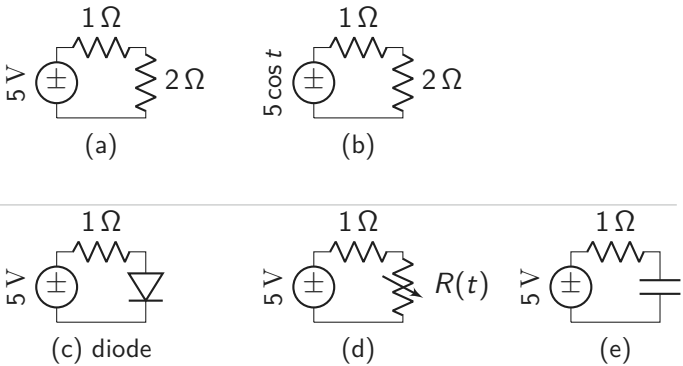
**Time-varying resistor**

$$v(t) = R(t)\, i(t)$$

the rule itself changes with $t$

Linear or nonlinear is also a property of the element law: $v = Ri$ is linear, a diode law is not.

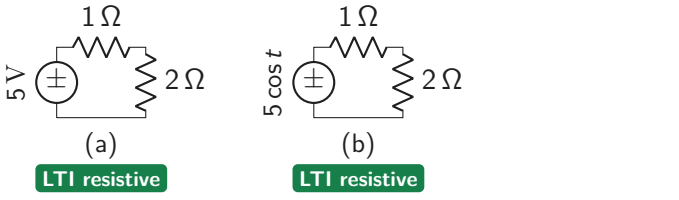
# Which circuit is which?




A source, a $1\,\Omega$ resistor and one more element. In (d), $R(t) = 2 + \sin t\ \Omega$.

Linear? Time-invariant? Resistive?

# Which circuit is which?




The input does not change the class: (a) and (b) are both LTI resistive.

Each of (c), (d), (e) breaks exactly one of the three properties.

## What we gain

**Always — no assumption on the elements**

KCL and KVL are linear for every circuit: they come from the graph alone.

# What we gain

## Always — no assumption on the elements

KCL and KVL are linear for every circuit: they come from the graph alone.

## Because the circuit is LTI resistive

Every element equation is linear and algebraic, with no derivative:

$$v = Ri \ \text{(resistor)}, \qquad v = \mu v_x, \quad i = g v_x, \ \ldots \ \text{(dependent sources)}$$

# What we gain

## Always — no assumption on the elements

KCL and KVL are linear for every circuit: they come from the graph alone.

## Because the circuit is LTI resistive

Every element equation is linear and algebraic, with no derivative:

$$v = Ri \quad \text{(resistor)}, \qquad v = \mu v_x, \quad i = g v_x, \quad \ldots \quad \text{(dependent sources)}$$

- Time enters only through the values of the independent sources.

## What we gain

### Always — no assumption on the elements

KCL and KVL are linear for every circuit: they come from the graph alone.

### Because the circuit is LTI resistive

Every element equation is linear and algebraic, with no derivative:

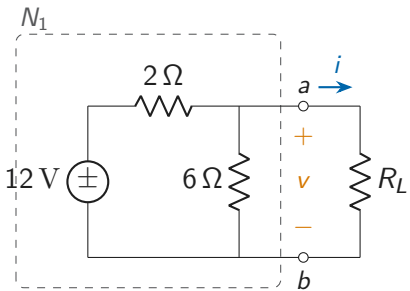
$$v = Ri \quad \text{(resistor)}, \qquad v = \mu v_x, \quad i = g v_x, \quad \ldots \quad \text{(dependent sources)}$$

- Time enters only through the values of the independent sources.
- So at each instant the circuit is solved like a circuit with constant sources: $2b$ linear algebraic equations.

# Roadmap

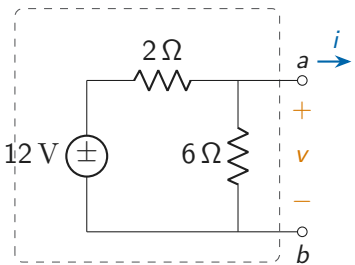
1. **LTI resistive circuits** — what the words mean, and what we gain ✓
2. **A first example** — simplify a circuit step by step
3. **Substitution theorem** — replace a part of a circuit by a source
4. **One-ports and equivalence** — the same $i$–$v$ characteristic at the port
5. **Series and parallel** — combine resistors and sources
6. **One source and one resistor** — exchange a voltage source and a current source

# A first example: what does $R_L$ see?




$R_L$ (the load) can be any resistive element.
Find $v$ and $i$.

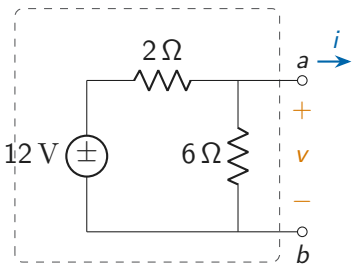
# A first example: what does $R_L$ see?




$R_L$ (the load) can be any resistive element. Find $v$ and $i$.

Leave $R_L$ aside: look only at the part left of $a$–$b$.

# A first example: what does $R_L$ see?

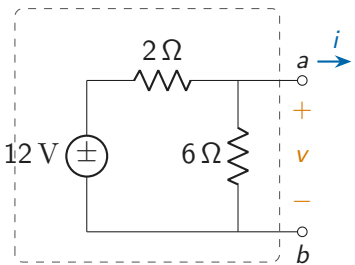



$R_L$ (the load) can be any resistive element. Find $v$ and $i$.

Leave $R_L$ aside: look only at the part left of $a$–$b$.

The $2\,\Omega$ has $12 - v$ across it, the $6\,\Omega$ has $v$.

# A first example: what does $R_L$ see?



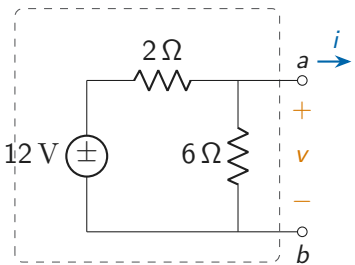

$R_L$ (the load) can be any resistive element. Find $v$ and $i$.

Leave $R_L$ aside: look only at the part left of $a$–$b$.

The $2\,\Omega$ has $12 - v$ across it, the $6\,\Omega$ has $v$.

KCL at $a$: $$\frac{12 - v}{2} = \frac{v}{6} + i$$

# A first example: what does $R_L$ see?




$R_L$ (the load) can be any resistive element. Find $v$ and $i$.

Leave $R_L$ aside: look only at the part left of $a$–$b$.

The $2\,\Omega$ has $12 - v$ across it, the $6\,\Omega$ has $v$.

KCL at $a$: $$\frac{12 - v}{2} = \frac{v}{6} + i$$

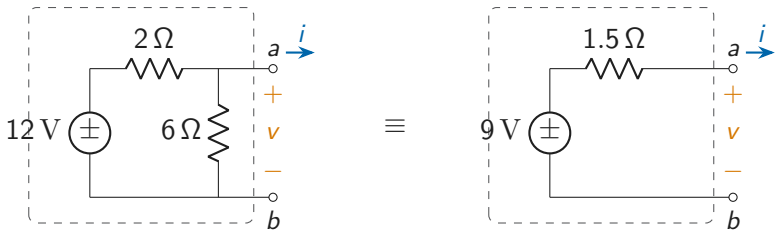
Solve for $v$: $v = 9 - 1.5\,i$

# Read the result as a circuit

$$v = 9 - 1.5\,i$$

# Read the result as a circuit

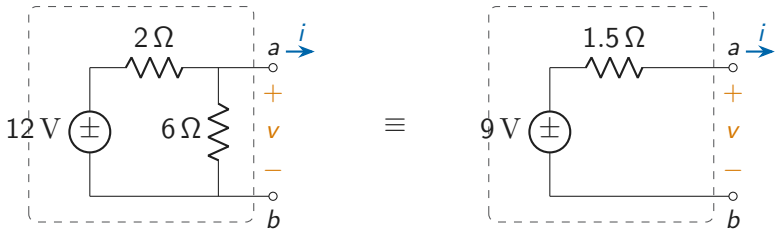
$$v = 9 - 1.5\,i$$




The right one obeys $v = 9 - 1.5\,i$ too (KVL): the same law.

# Read the result as a circuit

$$v = 9 - 1.5\,i$$




The right one obeys $v = 9 - 1.5\,i$ too (KVL): the same law.

Any load: $v = R_L i$, so $i = \dfrac{9}{1.5 + R_L}$. For $R_L = 3\,\Omega$: $i = 2\,\mathrm{A}$, $v = 6\,\mathrm{V}$.

## What did we just do?

- The part left of $a$–$b$ became one source and one resistor.

## What did we just do?

- The part left of $a$–$b$ became one source and one resistor.
- $R_L$ did not notice: every load gets the same $v$ and $i$ as before.

## What did we just do?

- The part left of $a$–$b$ became one source and one resistor.
- $R_L$ did not notice: every load gets the same $v$ and $i$ as before.
- The work is done once; each new load is one line of algebra.

# What did we just do?

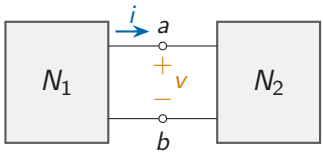
- The part left of $a$–$b$ became one source and one resistor.
- $R_L$ did not notice: every load gets the same $v$ and $i$ as before.
- The work is done once; each new load is one line of algebra.

| question | answered by |
|---|---|
| May a part be replaced by a source? | substitution theorem |
| When may a part be replaced by a simpler part? | equivalent one-ports |
| Which simple replacements are there? | series, parallel, source transformation |

# Roadmap

1. **LTI resistive circuits** — what the words mean, and what we gain ✓
2. **A first example** — simplify a circuit step by step ✓
3. **Substitution theorem** — replace a part of a circuit by a source
4. **One-ports and equivalence** — the same $i$–$v$ characteristic at the port
5. **Series and parallel** — combine resistors and sources
6. **One source and one resistor** — exchange a voltage source and a current source

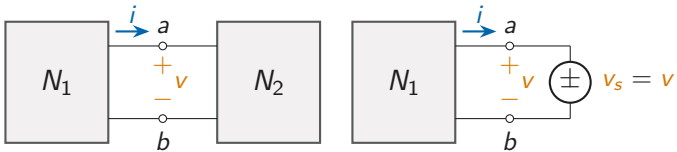
# The substitution theorem




(I)

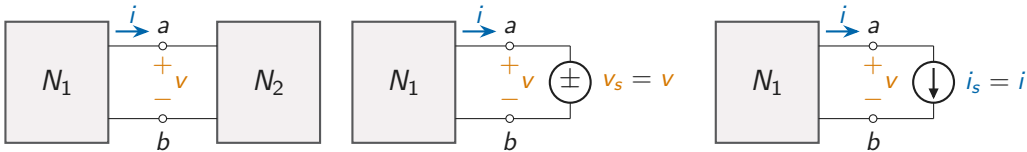
A circuit cut into two parts, joined at the terminals $a$, $b$.
$i$: current from $N_1$ into $N_2$ $v$: voltage between $a$ and $b$

# The substitution theorem



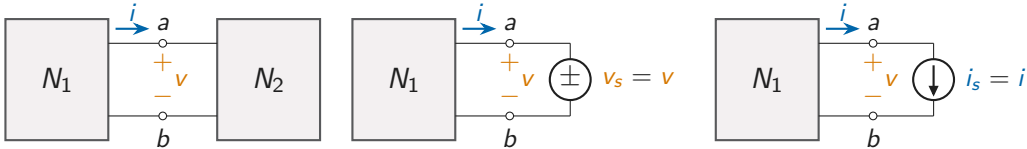

Replace $N_2$ by a voltage source with $v_s(t) = v(t)$ ...

# The substitution theorem




... or by a current source with $i_s(t) = i(t)$.

# The substitution theorem




## Substitution theorem

No branch voltage or branch current inside $N_1$ changes. An observer inside $N_1$ cannot tell (I), (II) and (III) apart.

Condition: each circuit has a unique solution.

# A unique solution

A circuit with $n$ nodes and $b$ branches has $2b$ unknowns, $v_1, \ldots, v_b$ and $i_1, \ldots, i_b$, and $2b$ equations:

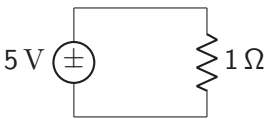
| | | |
|---|---|---|
| KCL | $A\mathbf{i} = 0$ | $n-1$ |
| KVL | $B\mathbf{v} = 0$ | $b-n+1$ |
| element laws | one for each branch | $b$ |

**Definition**

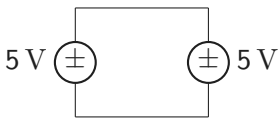
The circuit has a unique solution if exactly one $(\mathbf{v}, \mathbf{i})$ satisfies all $2b$ equations.

How do we check it? For an LTI resistive circuit: see the next slides.

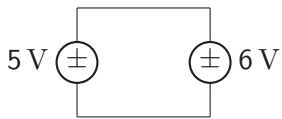
# A unique solution? Three small circuits




(a)

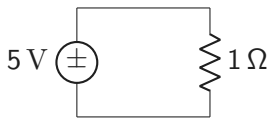



(b)



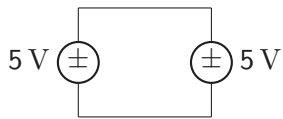

(c)

# A unique solution? Three small circuits



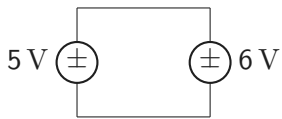

(a)

✓ unique: $i = 5\,\mathrm{A}$




(b)

× many: any current




(c)

× none: KVL fails

# A unique solution: in matrix form

Collect the $2b$ unknowns in one vector $x = (v_1, \ldots, v_b, i_1, \ldots, i_b)$. In an LTI resistive circuit all $2b$ equations are linear: $T\,x = u$.

$T$, the coefficient matrix: one row per equation, holding the numbers that multiply the unknowns. $u$: the source values.

The solution is unique exactly when $T$ is invertible: $\det T \neq 0$.

Branch 1: the left source; branch 2: the right element; both oriented from top to bottom.

# A unique solution: in matrix form

Collect the $2b$ unknowns in one vector $x = (v_1, \ldots, v_b, i_1, \ldots, i_b)$. In an LTI resistive circuit all $2b$ equations are linear: $T\,x = u$.

$T$, the coefficient matrix: one row per equation, holding the numbers that multiply the unknowns. $u$: the source values.

The solution is unique exactly when $T$ is invertible: $\det T \neq 0$.

(a) $5\,\text{V}$ ± $1\,\Omega$

$$\begin{bmatrix} 0 & 0 & 1 & 1 \\ 1 & -1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & -1 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \\ i_1 \\ i_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 5 \\ 0 \end{bmatrix}$$

KCL
KVL
source 1
resistor 2

$\det T = -1$: ✓ unique

Branch 1: the left source; branch 2: the right element; both oriented from top to bottom.

# A unique solution: in matrix form

Collect the $2b$ unknowns in one vector $x = (v_1, \ldots, v_b, i_1, \ldots, i_b)$. In an LTI resistive circuit all $2b$ equations are linear: $T\,x = u$.

$T$, the coefficient matrix: one row per equation, holding the numbers that multiply the unknowns. $u$: the source values.

The solution is unique exactly when $T$ is invertible: $\det T \neq 0$.

(a) $5\,\mathrm{V}$ ± — $1\,\Omega$

$$\begin{bmatrix} 0 & 0 & 1 & 1 \\ 1 & -1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & -1 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \\ i_1 \\ i_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 5 \\ 0 \end{bmatrix} \begin{matrix} \text{KCL} \\ \text{KVL} \\ \text{source 1} \\ \text{resistor 2} \end{matrix}$$

$\det T = -1$: ✓ unique

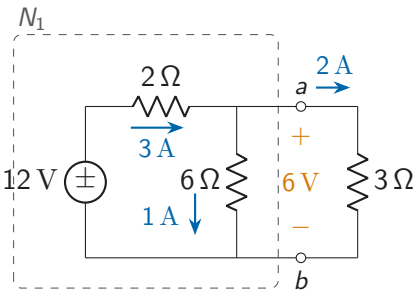
(b) $5\,\mathrm{V}$ ± — ± $5\,\mathrm{V}$

$$\begin{bmatrix} 0 & 0 & 1 & 1 \\ 1 & -1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \\ i_1 \\ i_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 5 \\ 5 \end{bmatrix} \begin{matrix} \text{KCL} \\ \text{KVL} \\ \text{source 1} \\ \text{source 2} \end{matrix}$$

$\det T = 0$: × $i_1$, $i_2$ appear only in KCL

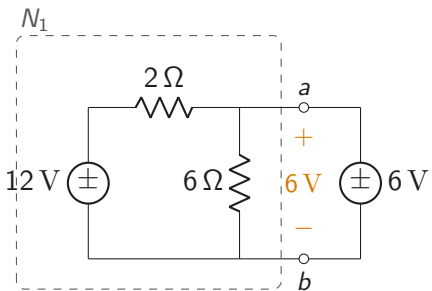
Branch 1: the left source; branch 2: the right element; both oriented from top to bottom.

# (I) and (II): an example




**(I)** The first example with $R_L = 3\,\Omega$:
$v = 6\,\mathrm{V}$, $i = 2\,\mathrm{A}$; $3\,\mathrm{A}$ in the $2\,\Omega$, $1\,\mathrm{A}$ in the $6\,\Omega$.

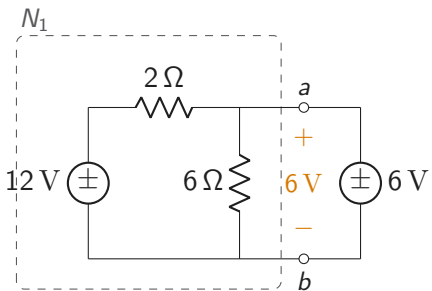
# (I) and (II): an example




**(I)** The first example with $R_L = 3\,\Omega$: $v = 6\,\mathrm{V}$, $i = 2\,\mathrm{A}$; $3\,\mathrm{A}$ in the $2\,\Omega$, $1\,\mathrm{A}$ in the $6\,\Omega$.

**(II)** Replace $R_L$ by a $6\,\mathrm{V}$ source. What are the values now?

# (I) and (II): an example

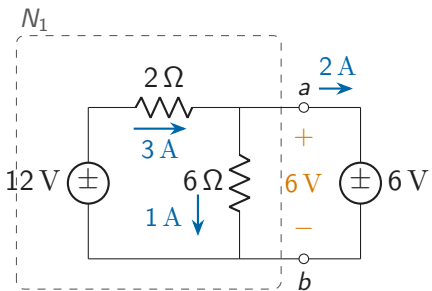



**(I)** The first example with $R_L = 3\,\Omega$:
$v = 6\,\mathrm{V}$, $i = 2\,\mathrm{A}$; $3\,\mathrm{A}$ in the $2\,\Omega$, $1\,\mathrm{A}$ in the $6\,\Omega$.

**(II)** Replace $R_L$ by a $6\,\mathrm{V}$ source. What are the values now?

KCL at $a$ with $v = 6\,\mathrm{V}$:
$$\frac{12-6}{2} = \frac{6}{6} + i \;\Rightarrow\; i = 2\,\mathrm{A}$$

# (I) and (II): an example




**(I)** The first example with $R_L = 3\,\Omega$: $v = 6\,\mathrm{V}$, $i = 2\,\mathrm{A}$; $3\,\mathrm{A}$ in the $2\,\Omega$, $1\,\mathrm{A}$ in the $6\,\Omega$.

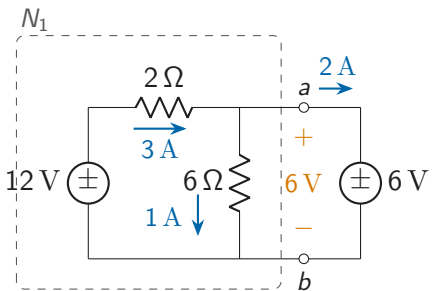
**(II)** Replace $R_L$ by a $6\,\mathrm{V}$ source. What are the values now?

KCL at $a$ with $v = 6\,\mathrm{V}$:

$$\frac{12-6}{2} = \frac{6}{6} + i \;\Rightarrow\; i = 2\,\mathrm{A}$$

$3\,\mathrm{A}$ in the $2\,\Omega$, $1\,\mathrm{A}$ in the $6\,\Omega$: nothing changed. ✓

# (I) and (II): an example




**(I)** The first example with $R_L = 3\,\Omega$: $v = 6\,\mathrm{V}$, $i = 2\,\mathrm{A}$; $3\,\mathrm{A}$ in the $2\,\Omega$, $1\,\mathrm{A}$ in the $6\,\Omega$.

**(II)** Replace $R_L$ by a $6\,\mathrm{V}$ source. What are the values now?

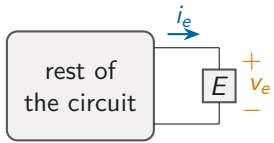
KCL at $a$ with $v = 6\,\mathrm{V}$:

$$\frac{12-6}{2} = \frac{6}{6} + i \;\Rightarrow\; i = 2\,\mathrm{A}$$

$3\,\mathrm{A}$ in the $2\,\Omega$, $1\,\mathrm{A}$ in the $6\,\Omega$: nothing changed. ✓

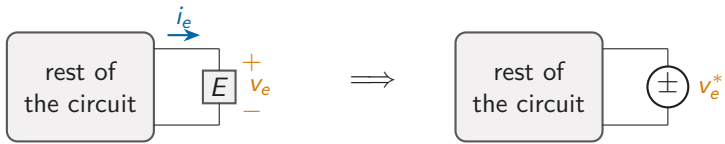
In (I) we solved the same equation: there too $v = 6\,\mathrm{V}$.

# (I) and (II): one element first



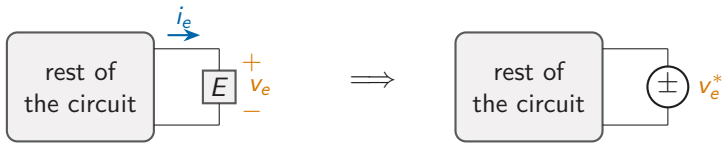

Any element $E$ of a circuit; in the solution of (I) it has $v_e^*$ and $i_e^*$.

# (I) and (II): one element first




Replace only $E$ by a voltage source. Assume both circuits have a unique solution.

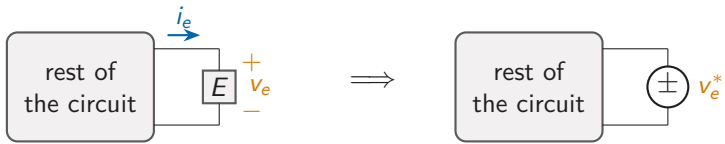
# (I) and (II): one element first




Replace only $E$ by a voltage source. Assume both circuits have a unique solution.

| | **(I)** |
|---|---|
| KCL | $A\,\mathbf{i} = 0$ |
| KVL | $B\,\mathbf{v} = 0$ |
| other $b-1$ elements | their laws |
| element $E$ | the law of $E$ |

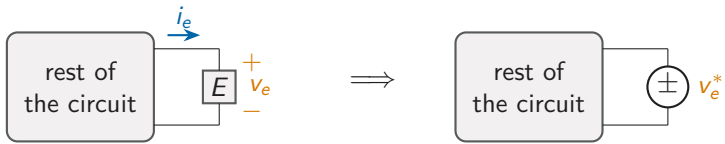
# (I) and (II): one element first




Replace only $E$ by a voltage source. Assume both circuits have a unique solution.

| | (I) | (II) |
|---|---|---|
| KCL | $A\,\mathbf{i} = 0$ | the same (same graph) |
| KVL | $B\,\mathbf{v} = 0$ | the same (same graph) |
| other $b-1$ elements | their laws | the same |
| element $E$ | the law of $E$ | $v_e = v_e^*$ |

# (I) and (II): one element first




Replace only $E$ by a voltage source. Assume both circuits have a unique solution.

| | (I) | (II) |
|---|---|---|
| KCL | $A\,\mathbf{i} = 0$ | the same (same graph) |
| KVL | $B\,\mathbf{v} = 0$ | the same (same graph) |
| other $b-1$ elements | their laws | the same |
| element $E$ | the law of $E$ | $v_e = v_e^*$ |

- The solution of (I), $(\mathbf{v}^*, \mathbf{i}^*)$, satisfies every equation of (I), the law of $E$ included.

# (I) and (II): one element, check the old solution

Does $(\mathbf{v}^*, \mathbf{i}^*)$ also satisfy the equations of (II)?

| | (I) | (II) |
|---|---|---|
| KCL | $A\mathbf{i} = 0$ | the same (same graph) |
| KVL | $B\mathbf{v} = 0$ | the same (same graph) |
| other $b-1$ elements | their laws | the same |
| element $E$ | the law of $E$ | $v_e = v_e^*$ |

# (I) and (II): one element, check the old solution

Does $(\mathbf{v}^*, \mathbf{i}^*)$ also satisfy the equations of (II)?

| | **(I)** | **(II)** | $(\mathbf{v}^*, \mathbf{i}^*)$ **in (II)?** |
|---|---|---|---|
| KCL | $A\,\mathbf{i} = 0$ | the same (same graph) | ✓ as in (I) |
| KVL | $B\,\mathbf{v} = 0$ | the same (same graph) | ✓ as in (I) |
| other $b-1$ elements | their laws | the same | ✓ as in (I) |
| element $E$ | the law of $E$ | $v_e = v_e^*$ | ✓ $v_e^* = v_e^*$ |

# (I) and (II): one element, check the old solution

Does $(\mathbf{v}^*, \mathbf{i}^*)$ also satisfy the equations of (II)?

| | (I) | (II) | $(\mathbf{v}^*, \mathbf{i}^*)$ in (II)? |
|---|---|---|---|
| KCL | $A\,\mathbf{i} = 0$ | the same (same graph) | ✓ as in (I) |
| KVL | $B\,\mathbf{v} = 0$ | the same (same graph) | ✓ as in (I) |
| other $b-1$ elements | their laws | the same | ✓ as in (I) |
| element $E$ | the law of $E$ | $v_e = v_e^*$ | ✓ $v_e^* = v_e^*$ |

- A voltage source takes any current, so $i_e = i_e^*$ is allowed.

# (I) and (II): one element, check the old solution

Does $(\mathbf{v}^*, \mathbf{i}^*)$ also satisfy the equations of (II)?

| | (I) | (II) | $(\mathbf{v}^*, \mathbf{i}^*)$ in (II)? |
|---|---|---|---|
| KCL | $A\,\mathbf{i} = 0$ | the same (same graph) | ✓ as in (I) |
| KVL | $B\,\mathbf{v} = 0$ | the same (same graph) | ✓ as in (I) |
| other $b-1$ elements | their laws | the same | ✓ as in (I) |
| element $E$ | the law of $E$ | $v_e = v_e^*$ | ✓ $v_e^* = v_e^*$ |

- A voltage source takes any current, so $i_e = i_e^*$ is allowed.
- So $(\mathbf{v}^*, \mathbf{i}^*)$ solves (II); it has only one solution, so nothing changes, also $i_e = i_e^*$.

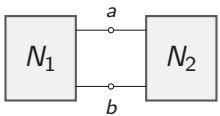
# (I) and (II): one element, check the old solution

Does $(\mathbf{v}^*, \mathbf{i}^*)$ also satisfy the equations of (II)?

| | (I) | (II) | $(\mathbf{v}^*, \mathbf{i}^*)$ in (II)? |
|---|---|---|---|
| KCL | $A\,\mathbf{i} = 0$ | the same (same graph) | ✓ as in (I) |
| KVL | $B\,\mathbf{v} = 0$ | the same (same graph) | ✓ as in (I) |
| other $b-1$ elements | their laws | the same | ✓ as in (I) |
| element $E$ | the law of $E$ | $v_e = v_e^*$ | ✓ $v_e^* = v_e^*$ |

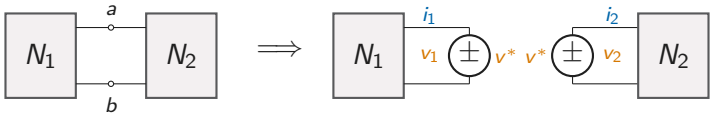
- A voltage source takes any current, so $i_e = i_e^*$ is allowed.
- So $(\mathbf{v}^*, \mathbf{i}^*)$ solves (II); it has only one solution, so nothing changes, also $i_e = i_e^*$.
- Uniqueness of (I) alone is not enough: in (a) of the three small circuits, replacing the $1\,\Omega$ by its voltage $5\,\mathrm{V}$ gives (b).

# (I) and (II): divide $N_1$ and $N_2$



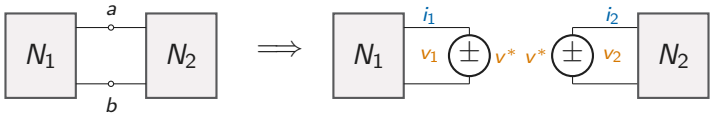

(I): $N_1$ and $N_2$ joined at $a$, $b$; the solution has $v^*$, $i^*$ at the port.

# (I) and (II): divide $N_1$ and $N_2$




Divide them and give each part its own source. Assume a unique solution.

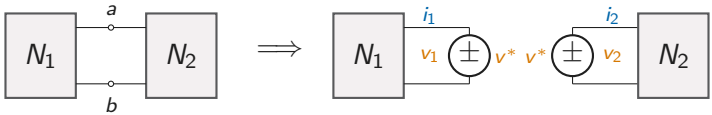
# (I) and (II): divide $N_1$ and $N_2$




Divide them and give each part its own source. Assume a unique solution.

| | **(I)** | **divided** |
|---|---|---|
| inside $N_1$ | KCL, KVL, element laws | the same |
| inside $N_2$ | KCL, KVL, element laws | the same |
| the connection | $v_1 = v_2,\ \ i_1 = i_2$ | $v_1 = v^*,\ \ v_2 = v^*$ |

# (I) and (II): divide $N_1$ and $N_2$

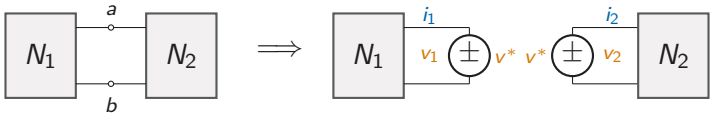



Divide them and give each part its own source. Assume a unique solution.

| | **(I)** | **divided** |
|---|---|---|
| inside $N_1$ | KCL, KVL, element laws | the same |
| inside $N_2$ | KCL, KVL, element laws | the same |
| the connection | $v_1 = v_2$, $i_1 = i_2$ | $v_1 = v^*$, $v_2 = v^*$ |

- The solution of (I) satisfies all these equations, since $v_1 = v_2 = v^*$. ✓

# (I) and (II): divide $N_1$ and $N_2$

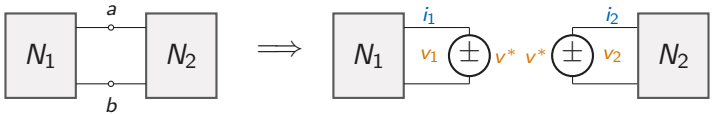



Divide them and give each part its own source. Assume a unique solution.

| | **(I)** | **divided** |
|---|---|---|
| inside $N_1$ | KCL, KVL, element laws | the same |
| inside $N_2$ | KCL, KVL, element laws | the same |
| the connection | $v_1 = v_2$, $i_1 = i_2$ | $v_1 = v^*$, $v_2 = v^*$ |

- The solution of (I) satisfies all these equations, since $v_1 = v_2 = v^*$. ✓
- One solution, so this is it: nothing changes in either part.

# (I) and (II): divide $N_1$ and $N_2$

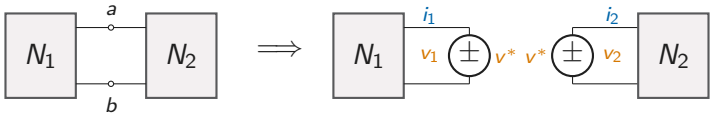



Divide them and give each part its own source. Assume a unique solution.

| | **(I)** | **divided** |
|---|---|---|
| inside $N_1$ | KCL, KVL, element laws | the same |
| inside $N_2$ | KCL, KVL, element laws | the same |
| the connection | $v_1 = v_2$, $i_1 = i_2$ | $v_1 = v^*$, $v_2 = v^*$ |

- The solution of (I) satisfies all these equations, since $v_1 = v_2 = v^*$. ✓
- One solution, so this is it: nothing changes in either part.
- The left part is exactly (II): nothing inside $N_1$ changes, also $i_1 = i^*$.

# (I) and (II): divide $N_1$ and $N_2$

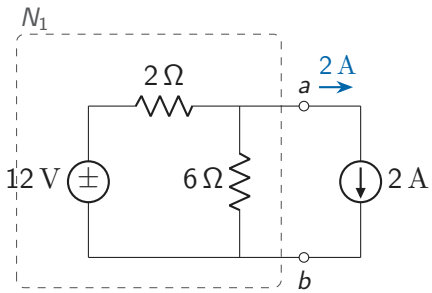



Divide them and give each part its own source. Assume a unique solution.

| | **(I)** | **divided** |
|---|---|---|
| inside $N_1$ | KCL, KVL, element laws | the same |
| inside $N_2$ | KCL, KVL, element laws | the same |
| the connection | $v_1 = v_2$, $i_1 = i_2$ | $v_1 = v^*$, $v_2 = v^*$ |

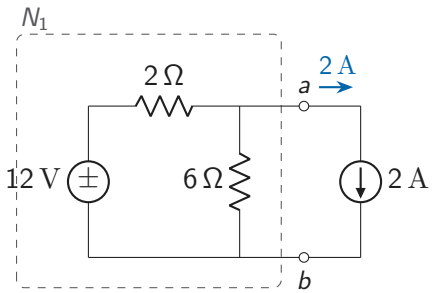
- The solution of (I) satisfies all these equations, since $v_1 = v_2 = v^*$. ✓
- One solution, so this is it: nothing changes in either part.
- The left part is exactly (II): nothing inside $N_1$ changes, also $i_1 = i^*$.
- $N_2$ is not needed at all: it can be anything.

# (I) and (III): an example




**(III)** Replace $R_L$ by a $2\,\mathrm{A}$ source. What are the values now?

# (I) and (III): an example

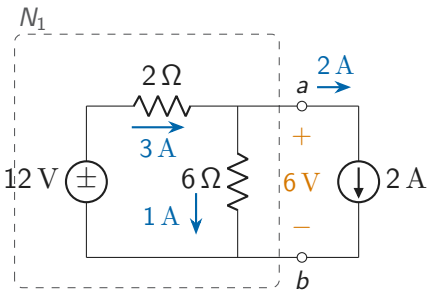



**(III)** Replace $R_L$ by a $2\,\mathrm{A}$ source. What are the values now?

KCL at $a$ with $i = 2\,\mathrm{A}$:

$$\frac{12 - v}{2} = \frac{v}{6} + 2 \;\Rightarrow\; v = 6\,\mathrm{V}$$

# (I) and (III): an example



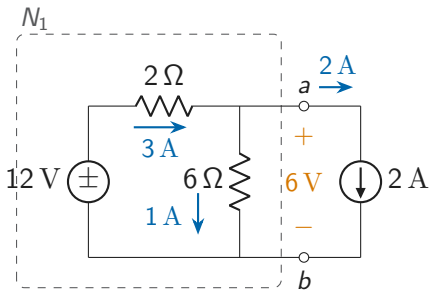

**(III)** Replace $R_L$ by a $2\,\mathrm{A}$ source. What are the values now?

KCL at $a$ with $i = 2\,\mathrm{A}$:

$$\frac{12 - v}{2} = \frac{v}{6} + 2 \;\Rightarrow\; v = 6\,\mathrm{V}$$

$3\,\mathrm{A}$ in the $2\,\Omega$, $1\,\mathrm{A}$ in the $6\,\Omega$: nothing changed. ✓

# (I) and (III): an example




**(III)** Replace $R_L$ by a $2\,\mathrm{A}$ source. What are the values now?

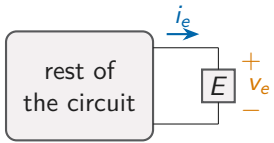
KCL at $a$ with $i = 2\,\mathrm{A}$:

$$\frac{12 - v}{2} = \frac{v}{6} + 2 \;\Rightarrow\; v = 6\,\mathrm{V}$$

$3\,\mathrm{A}$ in the $2\,\Omega$, $1\,\mathrm{A}$ in the $6\,\Omega$: nothing changed. ✓

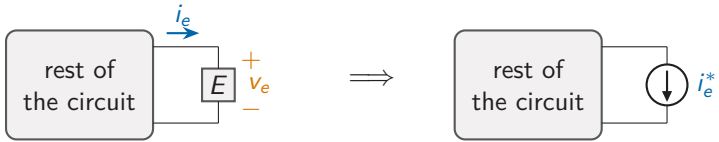
In (I) this equation holds too: there $i = 2\,\mathrm{A}$.

# (I) and (III): one element first



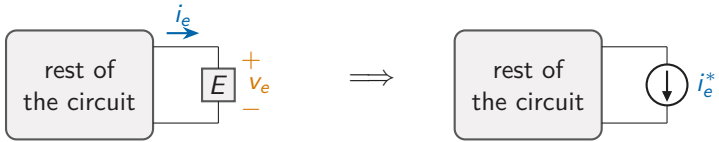

Any element $E$ of a circuit; in the solution of (I) it has $v_e^*$ and $i_e^*$.

# (I) and (III): one element first




Replace only $E$ by a current source. Assume both circuits have a unique solution.

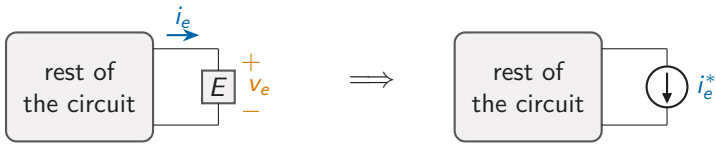
# (I) and (III): one element first




Replace only $E$ by a current source. Assume both circuits have a unique solution.

| | (I) |
|---|---|
| KCL | $A\,\mathbf{i} = 0$ |
| KVL | $B\,\mathbf{v} = 0$ |
| other $b-1$ elements | their laws |
| element $E$ | the law of $E$ |

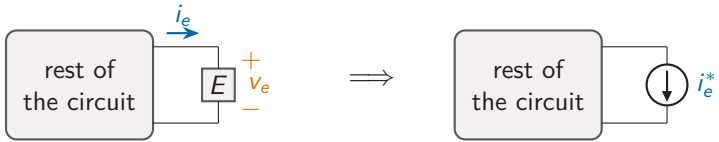
# (I) and (III): one element first




Replace only $E$ by a current source. Assume both circuits have a unique solution.

| | **(I)** | **(III)** |
|---|---|---|
| KCL | $A\,\mathbf{i} = 0$ | the same (same graph) |
| KVL | $B\,\mathbf{v} = 0$ | the same (same graph) |
| other $b-1$ elements | their laws | the same |
| element $E$ | the law of $E$ | $i_e = i_e^*$ |

# (I) and (III): one element first




Replace only $E$ by a current source. Assume both circuits have a unique solution.

| | (I) | (III) |
|---|---|---|
| KCL | $A\,\mathbf{i} = 0$ | the same (same graph) |
| KVL | $B\,\mathbf{v} = 0$ | the same (same graph) |
| other $b-1$ elements | their laws | the same |
| element $E$ | the law of $E$ | $i_e = i_e^*$ |

- The solution of (I), $(\mathbf{v}^*, \mathbf{i}^*)$, satisfies every equation of (I), the law of $E$ included.

# (I) and (III): one element, check the old solution

Does $(\mathbf{v}^*, \mathbf{i}^*)$ also satisfy the equations of (III)?

| | (I) | (III) |
|---|---|---|
| KCL | $A\,\mathbf{i} = 0$ | the same (same graph) |
| KVL | $B\,\mathbf{v} = 0$ | the same (same graph) |
| other $b-1$ elements | their laws | the same |
| element $E$ | the law of $E$ | $i_e = i_e^*$ |

# (I) and (III): one element, check the old solution

Does $(\mathbf{v}^*, \mathbf{i}^*)$ also satisfy the equations of (III)?

| | (I) | (III) | $(\mathbf{v}^*, \mathbf{i}^*)$ in (III)? |
|---|---|---|---|
| KCL | $A\,\mathbf{i} = 0$ | the same (same graph) | ✓ as in (I) |
| KVL | $B\,\mathbf{v} = 0$ | the same (same graph) | ✓ as in (I) |
| other $b-1$ elements | their laws | the same | ✓ as in (I) |
| element $E$ | the law of $E$ | $i_e = i_e^*$ | ✓ $i_e^* = i_e^*$ |

# (I) and (III): one element, check the old solution

Does $(\mathbf{v}^*, \mathbf{i}^*)$ also satisfy the equations of (III)?

| | (I) | (III) | $(\mathbf{v}^*, \mathbf{i}^*)$ in (III)? |
|---|---|---|---|
| KCL | $A\,\mathbf{i} = 0$ | the same (same graph) | ✓ as in (I) |
| KVL | $B\,\mathbf{v} = 0$ | the same (same graph) | ✓ as in (I) |
| other $b-1$ elements | their laws | the same | ✓ as in (I) |
| element $E$ | the law of $E$ | $i_e = i_e^*$ | ✓ $i_e^* = i_e^*$ |

- A current source takes any voltage, so $v_e = v_e^*$ is allowed.

# (I) and (III): one element, check the old solution

Does $(\mathbf{v}^*, \mathbf{i}^*)$ also satisfy the equations of (III)?

| | (I) | (III) | $(\mathbf{v}^*, \mathbf{i}^*)$ in (III)? |
|---|---|---|---|
| KCL | $A\,\mathbf{i} = 0$ | the same (same graph) | ✓ as in (I) |
| KVL | $B\,\mathbf{v} = 0$ | the same (same graph) | ✓ as in (I) |
| other $b-1$ elements | their laws | the same | ✓ as in (I) |
| element $E$ | the law of $E$ | $i_e = i_e^*$ | ✓ $i_e^* = i_e^*$ |

- A current source takes any voltage, so $v_e = v_e^*$ is allowed.
- So $(\mathbf{v}^*, \mathbf{i}^*)$ solves (III); it has only one solution, so nothing changes, also $v_e = v_e^*$.

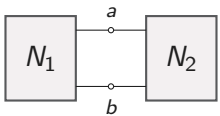
# (I) and (III): one element, check the old solution

Does $(\mathbf{v}^*, \mathbf{i}^*)$ also satisfy the equations of (III)?

| | (I) | (III) | $(\mathbf{v}^*, \mathbf{i}^*)$ in (III)? |
|---|---|---|---|
| KCL | $A\,\mathbf{i} = 0$ | the same (same graph) | ✓ as in (I) |
| KVL | $B\,\mathbf{v} = 0$ | the same (same graph) | ✓ as in (I) |
| other $b-1$ elements | their laws | the same | ✓ as in (I) |
| element $E$ | the law of $E$ | $i_e = i_e^*$ | ✓ $i_e^* = i_e^*$ |

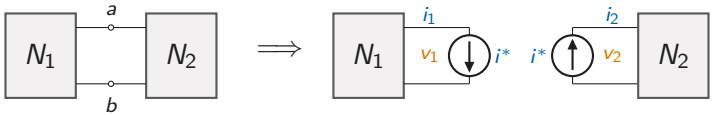
- A current source takes any voltage, so $v_e = v_e^*$ is allowed.
- So $(\mathbf{v}^*, \mathbf{i}^*)$ solves (III); it has only one solution, so nothing changes, also $v_e = v_e^*$.
- Uniqueness of (I) alone is not enough: a $2\,\mathrm{A}$ source with a $1\,\Omega$ becomes two $2\,\mathrm{A}$ sources in series.

# (I) and (III): divide $N_1$ and $N_2$



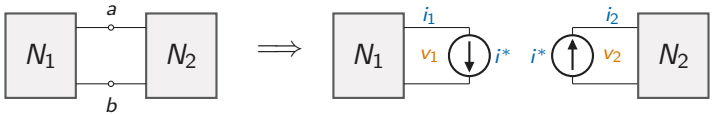

(I): $N_1$ and $N_2$ joined at $a$, $b$; the solution has $v^*$, $i^*$ at the port.

# (I) and (III): divide $N_1$ and $N_2$




Divide them and give each part its own source. Assume a unique solution.

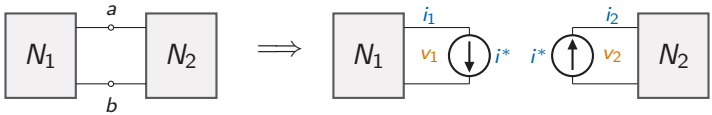
# (I) and (III): divide $N_1$ and $N_2$




Divide them and give each part its own source. Assume a unique solution.

| | **(I)** | **divided** |
|---|---|---|
| inside $N_1$ | KCL, KVL, element laws | the same |
| inside $N_2$ | KCL, KVL, element laws | the same |
| the connection | $v_1 = v_2, \;\; i_1 = i_2$ | $i_1 = i^*, \;\; i_2 = i^*$ |

# (I) and (III): divide $N_1$ and $N_2$

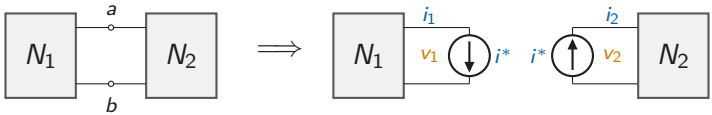



Divide them and give each part its own source. Assume a unique solution.

| | **(I)** | **divided** |
|---|---|---|
| inside $N_1$ | KCL, KVL, element laws | the same |
| inside $N_2$ | KCL, KVL, element laws | the same |
| the connection | $v_1 = v_2,\ \ i_1 = i_2$ | $i_1 = i^*,\ \ i_2 = i^*$ |

- The solution of (I) satisfies all these equations, since $i_1 = i_2 = i^*$. ✓

# (I) and (III): divide $N_1$ and $N_2$

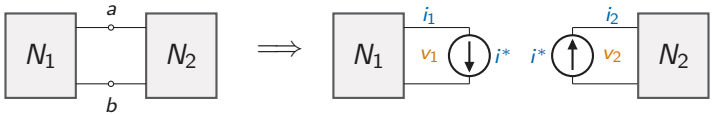



Divide them and give each part its own source. Assume a unique solution.

| | **(I)** | **divided** |
|---|---|---|
| inside $N_1$ | KCL, KVL, element laws | the same |
| inside $N_2$ | KCL, KVL, element laws | the same |
| the connection | $v_1 = v_2$, $i_1 = i_2$ | $i_1 = i^*$, $i_2 = i^*$ |

- The solution of (I) satisfies all these equations, since $i_1 = i_2 = i^*$. ✓
- One solution, so this is it: nothing changes in either part.

# (I) and (III): divide $N_1$ and $N_2$

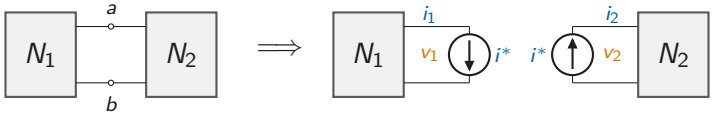



Divide them and give each part its own source. Assume a unique solution.

| | **(I)** | **divided** |
|---|---|---|
| inside $N_1$ | KCL, KVL, element laws | the same |
| inside $N_2$ | KCL, KVL, element laws | the same |
| the connection | $v_1 = v_2$, $i_1 = i_2$ | $i_1 = i^*$, $i_2 = i^*$ |

- The solution of (I) satisfies all these equations, since $i_1 = i_2 = i^*$. ✓
- One solution, so this is it: nothing changes in either part.
- The left part is exactly (III): nothing inside $N_1$ changes, also $v_1 = v^*$.

# (I) and (III): divide $N_1$ and $N_2$




Divide them and give each part its own source. Assume a unique solution.

| | **(I)** | **divided** |
|---|---|---|
| inside $N_1$ | KCL, KVL, element laws | the same |
| inside $N_2$ | KCL, KVL, element laws | the same |
| the connection | $v_1 = v_2$, $i_1 = i_2$ | $i_1 = i^*$, $i_2 = i^*$ |

- The solution of (I) satisfies all these equations, since $i_1 = i_2 = i^*$. ✓
- One solution, so this is it: nothing changes in either part.
- The left part is exactly (III): nothing inside $N_1$ changes, also $v_1 = v^*$.
- $N_2$ is not needed at all: it can be anything.

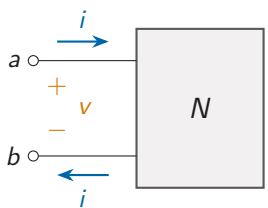
# Roadmap

1. **LTI resistive circuits** — what the words mean, and what we gain ✓
2. **A first example** — simplify a circuit step by step ✓
3. **Substitution theorem** — replace a part of a circuit by a source ✓
4. **One-ports and equivalence** — the same $i$–$v$ characteristic at the port
5. **Series and parallel** — combine resistors and sources
6. **One source and one resistor** — exchange a voltage source and a current source

# Ports and one-ports

## Port, one-port

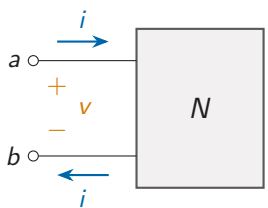
A pair of terminals is a port if the current entering one terminal equals the current leaving the other. A one-port is a circuit, or a circuit element, with one pair of terminals accessible from outside.



# Ports and one-ports

## Port, one-port

A pair of terminals is a port if the current entering one terminal equals the current leaving the other. A one-port is a circuit, or a circuit element, with one pair of terminals accessible from outside.

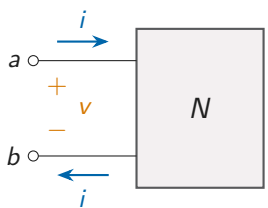



- A one-port is always a port: KCL on a closed surface around $N$.

# Ports and one-ports

## Port, one-port

A pair of terminals is a port if the current entering one terminal equals the current leaving the other. A one-port is a circuit, or a circuit element, with one pair of terminals accessible from outside.

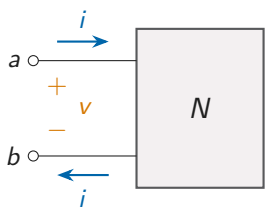



- A one-port is always a port: KCL on a closed surface around $N$.
- $i$ enters the $+$ terminal of $v$.

# Ports and one-ports

## Port, one-port

A pair of terminals is a port if the current entering one terminal equals the current leaving the other. A one-port is a circuit, or a circuit element, with one pair of terminals accessible from outside.

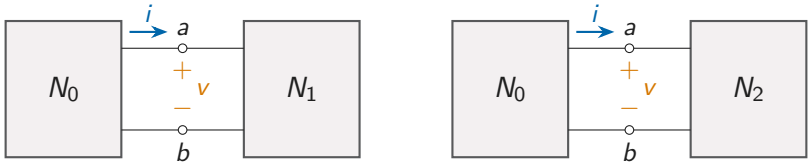



- A one-port is always a port: KCL on a closed surface around $N$.
- $i$ enters the $+$ terminal of $v$.
- The relation between $v$ and $i$ is the $i$–$v$ characteristic, e.g. $v = Ri$ for a resistor.

# Equivalent one-ports

## Equivalent one-ports

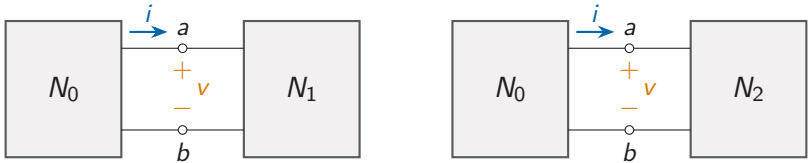
$N_1 \equiv N_2$ if they have the same $i$–$v$ characteristic at their terminals.



# Equivalent one-ports

## Equivalent one-ports

$N_1 \equiv N_2$ if they have the same $i$–$v$ characteristic at their terminals.

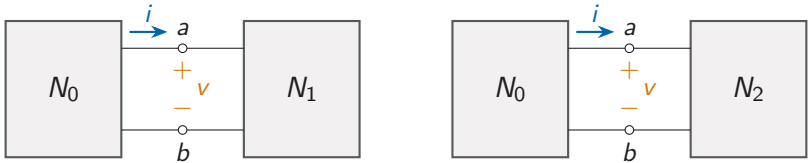



For any $N_0$: every branch voltage and current inside $N_0$ is the same in both.

# Equivalent one-ports

## Equivalent one-ports

$N_1 \equiv N_2$ if they have the same $i$–$v$ characteristic at their terminals.



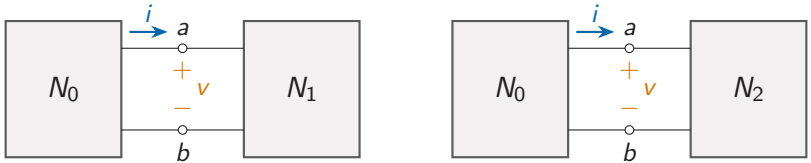

For any $N_0$: every branch voltage and current inside $N_0$ is the same in both.

Why: the equations are those of $N_0$ plus the $i$–$v$ characteristic at the port.

# Equivalent one-ports

## Equivalent one-ports

$N_1 \equiv N_2$ if they have the same $i$–$v$ characteristic at their terminals.




For any $N_0$: every branch voltage and current inside $N_0$ is the same in both.

Why: the equations are those of $N_0$ plus the $i$–$v$ characteristic at the port.

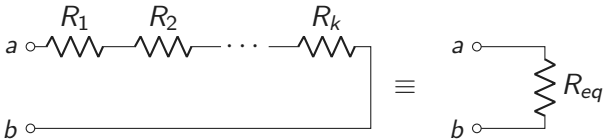
Inside $N_1$ and $N_2$ the branch values may differ.

# Roadmap

1. **LTI resistive circuits** — what the words mean, and what we gain ✓
2. **A first example** — simplify a circuit step by step ✓
3. **Substitution theorem** — replace a part of a circuit by a source ✓
4. **One-ports and equivalence** — the same $i$–$v$ characteristic at the port ✓
5. **Series and parallel** — combine resistors and sources
6. **One source and one resistor** — exchange a voltage source and a current source

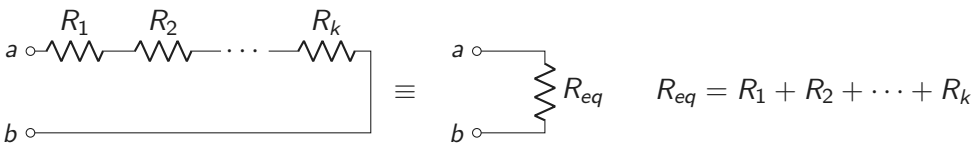
# Series connection

Same current $i$ through every element (KCL); the voltages add (KVL).



# Series connection

Same current $i$ through every element (KCL); the voltages add (KVL).

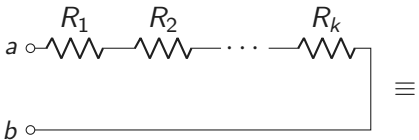



$$R_{eq} = R_1 + R_2 + \cdots + R_k$$

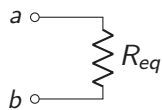
$$v = R_1 i + R_2 i + \cdots + R_k i = R_{eq} i$$

# Series connection

Same current $i$ through every element (KCL); the voltages add (KVL).




$\equiv$

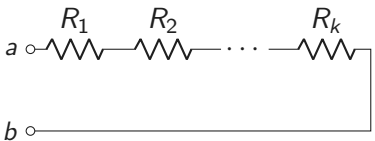



$$R_{eq} = R_1 + R_2 + \cdots + R_k$$

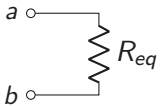
Voltage divider: the share of $R_j$ is $v_j = \dfrac{R_j}{R_{eq}}\, v$

# Series connection

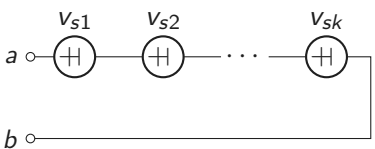
Same current $i$ through every element (KCL); the voltages add (KVL).



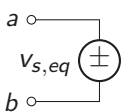

$\equiv$




$$R_{eq} = R_1 + R_2 + \cdots + R_k$$




$\equiv$

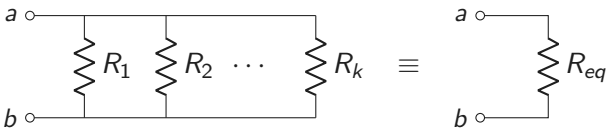



$$v_{s,eq} = v_{s1} + v_{s2} + \cdots + v_{sk}$$

All $+$ signs towards $a$; a source turned the other way enters with $-$.

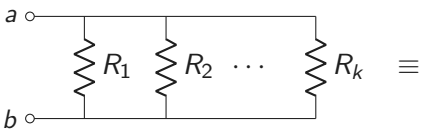
# Parallel connection

Same voltage $v$ across every element (KVL); the currents add (KCL).



# Parallel connection

Same voltage $v$ across every element (KVL); the currents add (KCL).

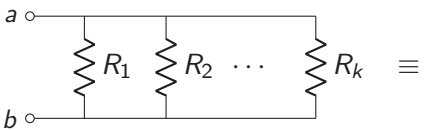



$$\frac{1}{R_{eq}} = \frac{1}{R_1} + \frac{1}{R_2} + \cdots + \frac{1}{R_k}$$

With $G_j = 1/R_j$: $\quad i = G_1 v + \cdots + G_k v = G_{eq} v, \quad G_{eq} = G_1 + \cdots + G_k$

# Parallel connection

Same voltage $v$ across every element (KVL); the currents add (KCL).

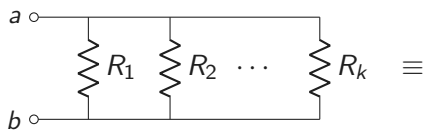



$$\frac{1}{R_{eq}} = \frac{1}{R_1} + \frac{1}{R_2} + \cdots + \frac{1}{R_k}$$

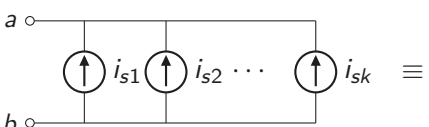
Current divider: $i_j = \dfrac{G_j}{G_{eq}}\, i.$ Two resistors: $R_{eq} = \dfrac{R_1 R_2}{R_1 + R_2}$

# Parallel connection

Same voltage $v$ across every element (KVL); the currents add (KCL).



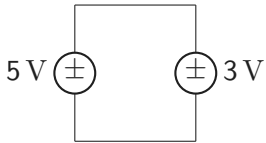

$$\frac{1}{R_{eq}} = \frac{1}{R_1} + \frac{1}{R_2} + \cdots + \frac{1}{R_k}$$



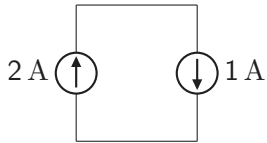

$$i_{s,eq} = i_{s1} + i_{s2} + \cdots + i_{sk}$$

All arrows towards $a$; a source turned the other way enters with $-$.

# Two connections to avoid



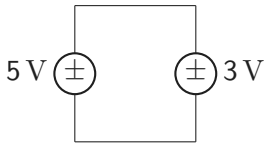

two voltage sources in parallel




two current sources in series

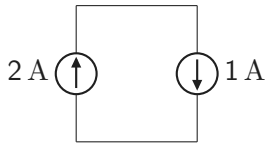
Are these allowed?

# Two connections to avoid




two voltage sources in parallel

KVL fails: $5 \neq 3$.




two current sources in series

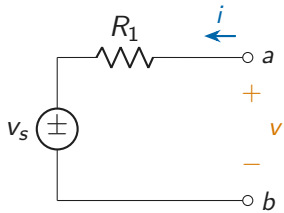
KCL fails: $2 \neq 1$.

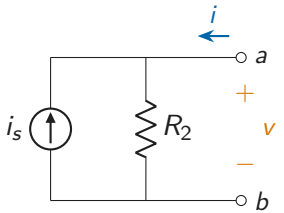
With equal values they are allowed, but the shared current (or voltage) is not fixed.

# Roadmap

1. **LTI resistive circuits** — what the words mean, and what we gain ✓
2. **A first example** — simplify a circuit step by step ✓
3. **Substitution theorem** — replace a part of a circuit by a source ✓
4. **One-ports and equivalence** — the same $i$–$v$ characteristic at the port ✓
5. **Series and parallel** — combine resistors and sources ✓
6. **One source and one resistor** — exchange a voltage source and a current source

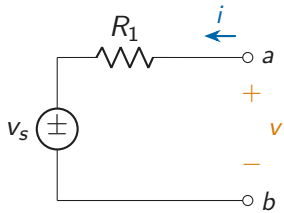
# One source and one resistor

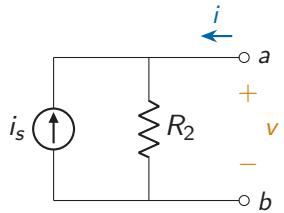



When do these two one-ports have the same $i$–$v$ characteristic?

# One source and one resistor

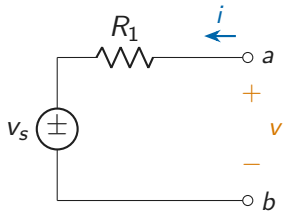


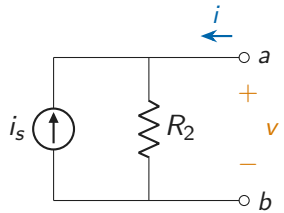

$$v = R_1 i + v_s$$

KVL around the loop: $R_1$ carries $i$.

# One source and one resistor



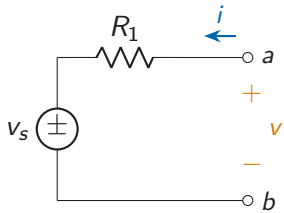

$v = R_1 i + v_s$



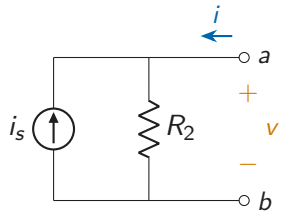

$v = R_2(i + i_s) = R_2 i + R_2 i_s$

$i$ and $i_s$ both flow into the top node: $R_2$ carries $i + i_s$.

# One source and one resistor




$$v = R_1 i + v_s$$




$$v = R_2(i + i_s) = R_2 i + R_2 i_s$$

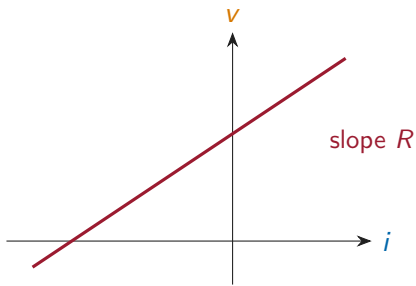
Both are straight lines in the $i$–$v$ plane: equal slopes and equal intercepts.

# Source transformation

## Source transformation

The two one-ports are equivalent if

$$R_1 = R_2 = R \qquad \text{and} \qquad v_s = R\, i_s.$$



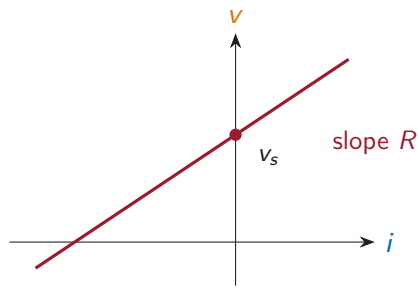
# Source transformation

## Source transformation

The two one-ports are equivalent if

$$R_1 = R_2 = R \qquad \text{and} \qquad v_s = R\, i_s.$$

Open circuit ($i = 0$): $v = v_s$.



# Source transformation

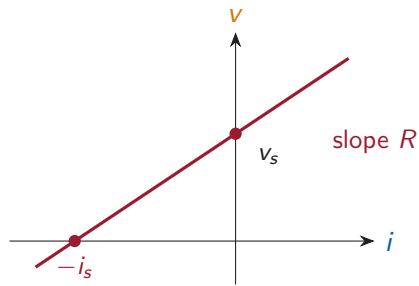
## Source transformation

The two one-ports are equivalent if

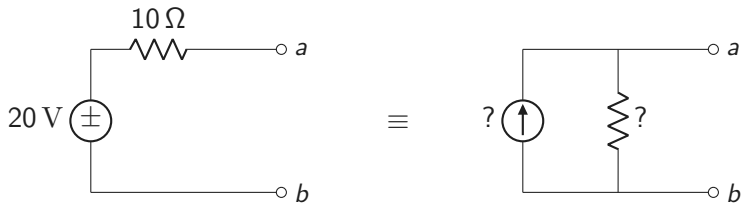
$$R_1 = R_2 = R \qquad \text{and} \qquad v_s = R\, i_s.$$

Open circuit ($i = 0$): $v = v_s$.

Short circuit ($v = 0$): $i = -i_s$.

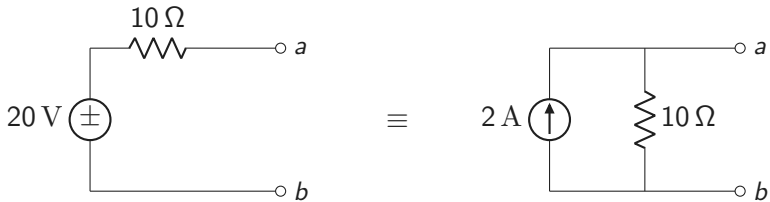


# Example: a source transformation




Find the current source and the resistor.

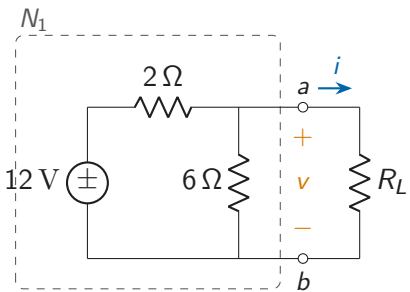
# Example: a source transformation




$$R = 10\,\Omega, \quad i_s = v_s/R = 20/10 = 2\,\mathrm{A}$$

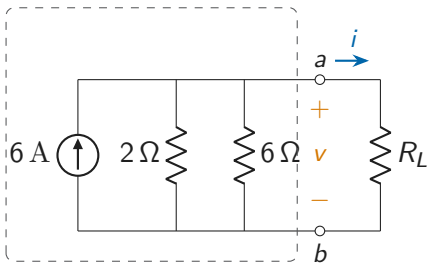
Both: open-circuit voltage $20\,\mathrm{V}$, short-circuit current $2\,\mathrm{A}$ (out of $a$).

# The first example, again



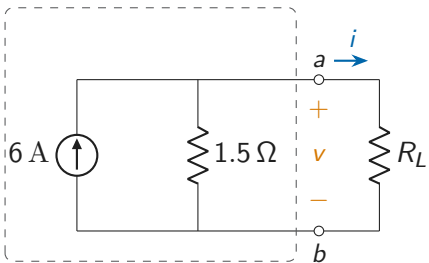

Now simplify only the part left of $a$–$b$; $R_L$ stays.

# The first example, again




**Step 1.** Source transformation:
$12\,\mathrm{V}$ with $2\,\Omega \rightarrow 6\,\mathrm{A}$ with $2\,\Omega$ $(12 = 2 \times 6)$

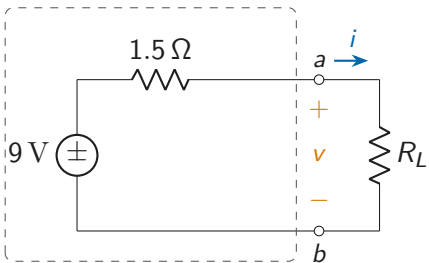
# The first example, again




**Step 1.** Source transformation:
$12\,\mathrm{V}$ with $2\,\Omega \rightarrow 6\,\mathrm{A}$ with $2\,\Omega$ $(12 = 2 \times 6)$

**Step 2.** Parallel: $2\,\Omega \parallel 6\,\Omega = 1.5\,\Omega$

# The first example, again

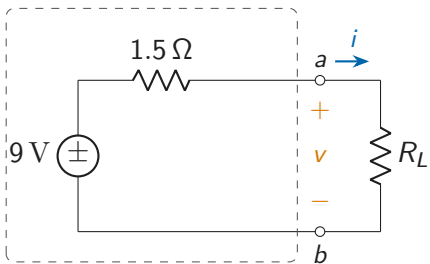



**Step 1.** Source transformation:
$12\,\mathrm{V}$ with $2\,\Omega \rightarrow 6\,\mathrm{A}$ with $2\,\Omega$ $(12 = 2 \times 6)$

**Step 2.** Parallel: $2\,\Omega \parallel 6\,\Omega = 1.5\,\Omega$

**Step 3.** Source transformation:
$6\,\mathrm{A}$ with $1.5\,\Omega \rightarrow 9\,\mathrm{V}$ with $1.5\,\Omega$
$(9 = 1.5 \times 6)$

# The first example, again



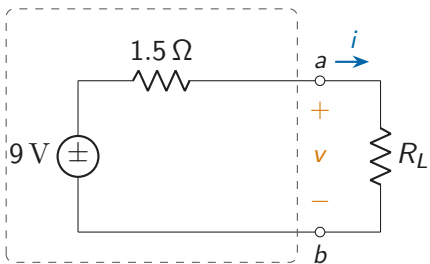

**Step 1.** Source transformation:
$12\,\mathrm{V}$ with $2\,\Omega \rightarrow 6\,\mathrm{A}$ with $2\,\Omega$ $(12 = 2 \times 6)$

**Step 2.** Parallel: $2\,\Omega \parallel 6\,\Omega = 1.5\,\Omega$

**Step 3.** Source transformation:
$6\,\mathrm{A}$ with $1.5\,\Omega \rightarrow 9\,\mathrm{V}$ with $1.5\,\Omega$
$(9 = 1.5 \times 6)$

Same $9\,\mathrm{V}$ and $1.5\,\Omega$, now by transforming the circuit.

# The first example, again




**Step 1.** Source transformation:
$12\,\mathrm{V}$ with $2\,\Omega \to 6\,\mathrm{A}$ with $2\,\Omega$ $(12 = 2 \times 6)$

**Step 2.** Parallel: $2\,\Omega \parallel 6\,\Omega = 1.5\,\Omega$

**Step 3.** Source transformation:
$6\,\mathrm{A}$ with $1.5\,\Omega \to 9\,\mathrm{V}$ with $1.5\,\Omega$
$(9 = 1.5 \times 6)$

Same $9\,\mathrm{V}$ and $1.5\,\Omega$, now by transforming the circuit.

Careful: inside, the values differ; for $R_L = 3\,\Omega$ the $12\,\mathrm{V}$ source carries $3\,\mathrm{A}$, the $9\,\mathrm{V}$ source $2\,\mathrm{A}$.

# Solving LTI resistive circuits

- Series and parallel connections and source transformations simplify many circuits, but not all.

# Solving LTI resistive circuits

- Series and parallel connections and source transformations simplify many circuits, but not all.
- Two methods write the equations of the circuit as it is, without transforming it, and work for every LTI resistive circuit:
  1. node voltage analysis: the $n-1$ node voltages are the unknowns;
  2. mesh current analysis: the $b-n+1$ mesh currents are the unknowns.

# Solving LTI resistive circuits

- Series and parallel connections and source transformations simplify many circuits, but not all.
- Two methods write the equations of the circuit as it is, without transforming it, and work for every LTI resistive circuit:
  1. node voltage analysis: the $n-1$ node voltages are the unknowns;
  2. mesh current analysis: the $b-n+1$ mesh currents are the unknowns.
- Both start from the matrices met earlier ($A$, $M$), now with linear element equations.

# Summary

- LTI resistive: apart from the independent sources (external inputs), every element is linear, time-invariant and memoryless, so all equations are linear and algebraic.
- Substitution: replace $N_2$ by a voltage source $v_s = v$ or a current source $i_s = i$; if each circuit has a unique solution, nothing changes inside $N_1$.
- Equivalent one-ports: the same $i$–$v$ characteristic; the rest cannot tell them apart.
- Series: $R_{eq} = \sum R_j$, $v_{s,eq} = \sum v_{sj}$. Parallel: $G_{eq} = \sum G_j$, $i_{s,eq} = \sum i_{sj}$.
- Source transformation: $v_s$ with series $R \equiv i_s = v_s/R$ with parallel $R$.